TECHRISE 3.0

Cybersecurity & DevSecOps Programme · Abia Cohort 3.0 · 2026

PHASE 2 · BLUE TEAM & DEFENCE

WEEK 06

SIEM Implementation and Detection Engineering

Student Training Manual — AI-First Curriculum

PROGRAMME	COHORT	WEEK	PHASE	METHODOLOGY
Cybersecurity & DevSecOps	**Abia Cohort 3.0**	**6 of 12**	**2 — Blue Team & Defence**	**AI-First (Manual → AI)**

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Phase: Blue Team and Defence (Weeks 5–8) **Week 6 of 12**

WEEK 6: SIEM IMPLEMENTATION AND DETECTION ENGINEERING

Weekly Overview

This week you transform from understanding attacks to detecting them at scale. You will learn the blue team mindset, deploy a SIEM (Splunk Free), ingest logs from your DVWA web server, and write detection rules that identify SQL injection, XSS, and brute force attacks. You will then enhance these manual detections with AI-powered anomaly detection. By the end of this week, you will have a functioning security monitoring lab and the skills to detect the attacks you learned in Week 5.

How this week builds on Weeks 1–5

- Week 3 taught you log analysis with grep/awk. Week 5 taught you web attacks (SQLi, XSS). Now you combine them: you will write SIEM queries (similar to grep pipelines) to detect those attacks in web logs.
- Your Week 2 risk assessment knowledge helps you prioritise which detections to build first.

How this week sets up Week 7

- Week 7 is IDS/IPS (Snort). SIEM detects attacks from logs; IDS detects from network traffic. Together they form a complete detection stack. Week 8 (Forensics) will use SIEM logs as evidence.

Why this matters in a real security job

- Every SOC (Security Operations Center) uses a SIEM. On average, SOC analysts spend 60% of their time in Splunk or similar tools. Without SIEM skills, you cannot work in a modern SOC.

■ Authorisation Reminder

All logs analysed this week are from your own lab environment (DVWA, Metasploitable, or your own systems). Never analyse logs from systems you do not own.

DAY
1

Blue Team Mindset and SOC Operations

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain the difference between red team and blue team mindsets

2. Describe SOC structure (Tier 1, 2, 3) and analyst responsibilities

3. Map the cyber kill chain to defensive controls

BLUE TEAM

is the group responsible for defending an organisation's information systems by monitoring, detecting, and responding to security incidents.

Red Team vs Blue Team Mindset

■■■

Phase	Attacker Action	Defensive Control	Detection Opportunity
1. Reconnaissance	Researching targets	Public info monitoring	Unusual DNS queries, WHOIS lookups
2. Weaponisation	Creating malware	Threat intelligence feeds	IoC feeds, malware sandboxes
3. Delivery	Phishing email / exploit	Email filtering, patching	Email gateway logs, IDS alerts
4. Exploitation	Running exploit code	EDR, application hardening	EDR alerts, application logs
5. Installation	Installing backdoor	AV, application whitelisting	File system monitoring, SIEM
6. C2	Communicating with attacker	Firewall, DNS filtering	Unusual outbound connections
7. Actions on Objectives	Stealing data	DLP, segmentation	Data exfiltration alerts, SIEM

Key SOC Metrics

Metric	Definition	Industry Benchmark	What It Tells You
MTTD	Mean Time to Detect — avg time from breach to detection	197 days (IBM 2023)	How quickly you find attackers in your environment
MTTR	Mean Time to Respond — avg time to contain after detection	69 days (IBM 2023)	How quickly you stop the bleeding after detection
False Positive Rate	% of alerts that are not real incidents	<20% is good	High false positives = alert fatigue = missed real attacks
Alert Coverage	% of MITRE ATT&CK; techniques with detection rules	Target: >60%	How much of the attack landscape you can see

WHY THIS MATTERS

The average dwell time (time attackers spend in a network before detection) is 197 days according to IBM's 2023 Cost of a Data Breach Report. Every day an attacker goes undetected costs an organisation approximately $5,500. A SOC that reduces MTTD from 197 days to 30 days saves approximately $920,000 per incident. SIEM is the primary tool for reducing MTTD.

PRACTICE *(2 Hours)*

LAB 6.1.1 — Blue Team Mindset Exercise

Objective: Apply the blue team mindset to analyse a simulated attack scenario.

Prerequisites: Week 4 knowledge of Metasploitable2 attack; Week 5 knowledge of SQLi/XSS

Review your Week 4 attack chain against Metasploitable2 and your Week 5 SQLi/XSS attacks. For each attack, answer the following from a blue team perspective:

1. What log file would record this attack? (auth.log, Apache access.log, syslog, etc.)
2. What specific log entry pattern would indicate this attack?
3. At which Kill Chain phase would a SOC analyst first have visibility?

4. What is the MTTD if the SOC only checks logs daily? What if they have a SIEM with real-time alerting?
5. What Tier 1 or Tier 2 action should be taken when this alert fires?

LAB 6.1.2 — SOC Alert Triage Simulation

Objective: Practice prioritising and triaging a queue of security alerts.

Prerequisites: No tools required — critical thinking exercise

You are a Tier 1 SOC analyst starting a shift. You have 8 alerts in your queue. Prioritise them (1=highest) and explain your escalation decision:

Alert	Severity	Your Priority	Escalate?	Reason
10 failed SSH logins from 192.168.1.100	Medium			
Successful login from 192.168.1.100 after 10 failures	High			
SQL injection pattern in Apache log from external IP	High			
Antivirus detected and quarantined trojan on finance PC	Critical			
User account created at 3:47 AM by admin account	High			
Port scan detected from internal IP 10.0.0.5	Medium			
200 failed login attempts on webmail from 185.x.x.x	Medium			
Outbound connection to known C2 IP from marketing PC	Critical			

■ ANALYSE THIS

Alert 2 (successful login after failures) is often called an 'attack chain' alert. Why is this more serious than either alert alone? What additional information would you need before escalating to Tier 2? What is the 'attack chain' connecting alerts 1, 2, and 3 if all from the same IP?

BUILD *(2 Hours)*

BUILD BRIEF

Create a SOC Alert Triage Playbook for a Tier 1 analyst.

Purpose:

- A new Tier 1 analyst on their first day should be able to use this document to triage all common alert types

Required sections:

- Alert Priority Framework: 4-level system (Critical/High/Medium/Low) with definitions and response times

- Decision Tree: flowchart for each alert type — when to close, escalate to Tier 2, or investigate further
- Common Alert Types: for each of the 8 alerts from Lab 6.1.2 — description, typical cause (true/false positive), investigation steps, escalation criteria
- Communication Template: how to write an escalation note to Tier 2 (format: Alert ID, What I found, What I did, Why I'm escalating)
- Shift Handover Template: what to document for the next analyst

Format:

- Professional operational document. Include a quick-reference card (1 page) that summarises the decision tree.

DOCUMENT

Writing Prompt:

Write a business case memo titled 'Why We Need a SIEM: The Business Case for Security Monitoring.' Audience: CEO and CFO. Length: 400–500 words.

- Required: Current state (no SIEM) risk exposure, cost of a breach vs cost of a SIEM (use IBM 2023 data), specific scenarios of what a SIEM would detect, ROI calculation
- Tone: Executive — focused on business risk and financial impact, not technical details
- Conclude with: recommended SIEM solution (Splunk Free for start), implementation timeline, expected MTTD improvement

DAY 2

SIEM Concepts and Splunk Free Installation

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain SIEM architecture and how log ingestion works

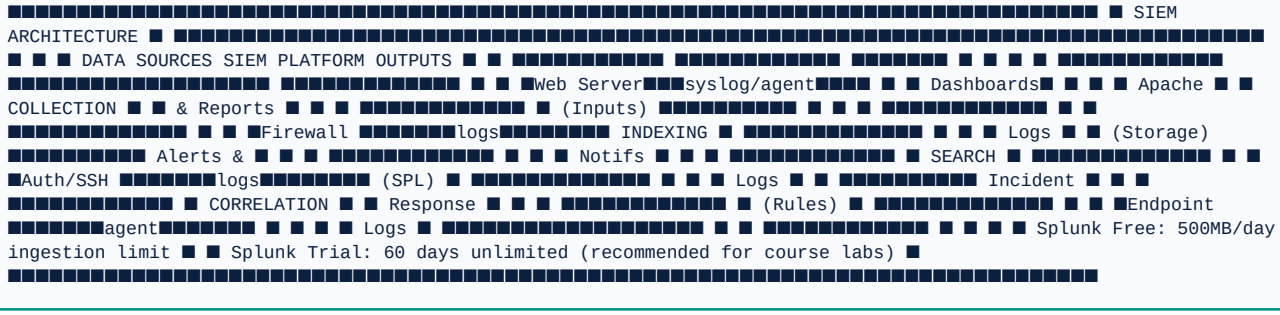
2. Install and configure Splunk Free on Kali Linux

3. Ingest DVWA Apache logs and verify data is searchable

SIEM (SECURITY INFORMATION AND EVENT MANAGEMENT)

is a platform that aggregates log data from multiple sources, normalises it into a searchable format, and enables real-time security monitoring, alerting, and incident investigation.

SIEM Architecture



Splunk Core Concepts

Concept	Definition	Security Analogy
Index	A repository where data is stored and made searchable	Like a filing cabinet for your logs
Source	The original file path the data came from	/var/log/apache2/access.log
Sourcetype	The format/type of the data	apache_access, linux_secure, syslog
Event	A single log entry	One line from your access.log
Field	A key-value pair extracted from an event	clientip=192.168.1.100, status=200
Search	A SPL query that finds and analyses events	Like grep but with statistics
Alert	A saved search that fires a notification when triggered	Like an automated grep that pages you

Dashboard	A collection of visualisations from saved searches	Your SOC monitoring screen

Splunk vs Wazuh — Comparison

Feature	Splunk Free	Wazuh
Cost	Free (500MB/day limit)	Completely free, open source
Data Limit	500MB/day (upgradeable)	Unlimited
Ease of Setup	Easy — single installer	Moderate — manager + agents
Search Language	SPL (powerful, flexible)	Kibana query language
Alerting	Built-in	Built-in with email/Slack
Built-in Rules	Limited in free version	3,000+ out-of-the-box SIEM rules
Agent Support	Universal Forwarder	Wazuh agents (Windows/Linux/Mac)
Best For	Learning SPL; enterprise SOC skills	Production HIDS; free SOC platform
Compliance	Paid version	PCI-DSS, HIPAA, GDPR built-in

WHY THIS MATTERS

Splunk commands 32% of the global SIEM market. In a survey of security job postings in 2023, 78% that required SIEM skills specifically mentioned Splunk. Learning SPL (Splunk Processing Language) is one of the highest-value skills for a SOC career — it appears in more job descriptions than any other security tool skill.

PRACTICE *(2 Hours)*

LAB 6.2.1 — Install and Configure Splunk Free

Objective: Deploy Splunk Free on Kali Linux and ingest your first log data.

Prerequisites: Kali Linux running; DVWA running; internet connection for download

Step 1 — Download Splunk Free

BASH

```
# Download Splunk Free (register at splunk.com first) wget -O splunk-9.x.x.tgz 'https://download.splunk.com/products/splunk/releases/[version]/linux/splunk-[version]-Linux-x86_64.tgz' # Or download .deb package wget -O splunk.deb 'https://download.splunk.com/products/splunk/releases/[version]/linux/splunk-[version]-linux-amd64.deb'
```

Step 2 — Install Splunk

BASH

```
# Install from .deb sudo dpkg -i splunk.deb # Start Splunk for the first time (sets admin password) sudo /opt/splunk/bin/splunk start --accept-license # Enable Splunk to start on boot sudo /opt/splunk/bin/splunk enable boot-start
```

Step 3 — Access Splunk Web Interface

Open browser → http://localhost:8000. Login with admin and the password you set. You should see the Splunk home page.

Step 4 — Add DVWA Apache Logs as Data Source

In Splunk: Settings → Data Inputs → Files & Directories → Add New.

File/Directory: /var/log/apache2/access.log. Source type: access_combined. Index: Create new index called 'dvwa'.

Step 5 — Verify Data Ingestion

SPL

```
index=dvwa | head 10
```

Expected: 10 most recent Apache log events. If you see results, Splunk is ingesting your DVWA logs correctly.

Step 6 — Also Add auth.log

Repeat the process for /var/log/auth.log with sourcetype=linux_secure and add to the same 'dvwa' index. This gives you both web and auth logs in one place.

■ ANALYSE THIS

Splunk shows your Apache log events with fields: clientip, method, uri_path, status, bytes. How are these fields extracted from the raw log line? What is the security value of field extraction — why is it better to search on status=500 than to grep for '500'?

BUILD *(2 Hours)*

BUILD BRIEF

Build your Splunk Lab Environment Documentation.

Required sections:

- Installation Log: exact commands run, any errors encountered and how resolved
- Data Sources Configured: for each source — file path, sourcetype, index, sample event
- Field Extraction Verification: run index=dvwa | table clientip, method, uri_path, status | head 20 — paste output
- Volume Baseline: index=dvwa | stats count by date_hour — how many events per hour?
- Data Quality Check: are there any blank fields? Malformed events? How would you fix them?

This document becomes your reference for the rest of Week 6 and the Checkpoint Project

DOCUMENT

Writing Prompt:

Write a SIEM Implementation Plan for a fictional Nigerian bank — 'First Digital Bank Nigeria.'

- Required: Current state assessment (what logs exist, where they are), SIEM selection justification (Splunk vs Wazuh vs other), data sources to ingest (priority order), staffing requirements (how many SOC analysts?), implementation timeline (30/60/90 days), compliance alignment (NDPA, CBN requirements)
- Budget-conscious: First Digital Bank is a startup — recommend the most cost-effective path to a functional SOC

DAY 3

Splunk Search Language (SPL) and Detection Rules

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Write SPL queries to search, filter, and aggregate security events

2. Create detection rules for SQL injection, XSS, and brute force attacks

3. Build a Splunk dashboard with real-time security visualisations

SPL (SPLUNK PROCESSING LANGUAGE)

is Splunk's query language, used to search and analyse indexed data. It uses a pipeline syntax (|) similar to Unix command chaining, where each command transforms the data flowing through the pipeline.

SPL Pipeline — How It Works

```
■■■
```

alert	Save search as alert	(configured in UI, not SPL)

Detection Rule Library — Web Attack Detection

SPL — SQL INJECTION DETECTION

```
index=dvwa sourcetype=access_combined | rex field=uri_path
"(?i)(?Punion|select|insert|drop|update|delete|'|--)" | where isnotnull(sqli_pattern) | stats count
by clientip, uri_path, sqli_pattern | sort -count | where count > 0
```

SPL — BRUTE FORCE DETECTION (SSH)

```
index=dvwa sourcetype=linux_secure "Failed password" | stats count by src_ip | where count > 10 | sort
-count
```

SPL — SUCCESSFUL LOGIN AFTER FAILURES (ATTACK CHAIN)

```
index=dvwa sourcetype=linux_secure | eval event_type=if(match(_raw,"Failed password"),"failure",
if(match(_raw,"Accepted password"),"success",null())) | where isnotnull(event_type) | stats
count(eval(event_type="failure")) as failures, count(eval(event_type="success")) as successes by src
| where failures > 5 AND successes > 0 | eval attack_chain="ALERT: Brute force followed by success"
```

SPL — HIGH ERROR RATE (APPLICATION ATTACK)

```
index=dvwa sourcetype=access_combined | stats count(eval(status>=400)) as errors, count as total by
clientip | eval error_rate=round(errors/total*100,1) | where error_rate > 50 AND total > 20 | sort
-error_rate
```

SPL — XSS DETECTION IN URL

```
index=dvwa sourcetype=access_combined | rex field=uri_path
"(?i)(?Pscript|onerror|onload|alert|document\.cookie)" | where isnotnull(xss_pattern) | stats count
by clientip, uri_path, xss_pattern | sort -count
```

WHY THIS MATTERS

In 2020, the Twitter hack that compromised accounts of Barack Obama, Elon Musk, and others for a Bitcoin scam was not detected by Twitter's SIEM until the attackers had already taken action. Post-incident analysis showed that the admin tool access logs contained clear anomalies — but no detection rule existed for admin access from an unknown IP. Building detection rules before an attack is what separates mature SOCs from reactive ones.

PRACTICE *(2 Hours)*

LAB 6.3.1 — Write and Test SPL Detection Rules

Objective: Build detection rules for attacks you performed in Weeks 4 and 5.

Prerequisites: Splunk running with dvwa index; DVWA running; run some attacks first

Step 1 — Generate Attack Traffic to Detect

Before writing detection rules, generate attack traffic so you have data to detect:

BASH

```
# Generate failed SSH logins for i in {1..15}; do ssh invalid_user@localhost; done # Run SQLi against DVWA (from Week 5) # Open DVWA → SQL Injection → enter: 1' OR '1'='1 # Run directory brute-force against DVWA gobuster dir -u http://localhost/dvwa -w /usr/share/wordlists/dirb/common.txt
```

Step 2 — Test SQL Injection Detection

SPL

```
index=dvwa sourcetype=access_combined | rex field=uri_path "(?i)(?Punion|select|'|--)" | where isnotnull(sqli) | table _time, clientip, uri_path, sqli, status
```

Expected: rows showing your SQLi attempts with the extracted pattern. If no results — verify DVWA logging is enabled and attack traffic was generated.

Step 3 — Test Brute Force Detection

SPL

```
index=dvwa sourcetype=linux_secure "Failed password" | stats count by src_ip | where count > 5 | sort -count
```

Step 4 — Create a Splunk Alert

Take your SQL injection detection query. Save it as an Alert: Save As → Alert. Name: SQLi_Detection. Alert type: Real-time. Trigger condition: Number of results is greater than 0. Action: Log event (free version). Review the alert in Alerts tab.

Step 5 — Build a Security Dashboard

Create a new Dashboard: Dashboards → Create New Dashboard → Classic Dashboard. Add panels: (1) SQL Injection Attempts (table), (2) Failed Logins by IP (bar chart), (3) HTTP Status Code Distribution (pie chart), (4) Top Attacking IPs (table). Save and set to auto-refresh every 60 seconds.

■ ANALYSE THIS

Your SQL injection detection rule fires on the string 'select' in the URL. This would also fire if a legitimate user searched for a product called 'SELECT Coffee' on a web store. How would you reduce false positives while keeping true positive detection? What additional conditions would you add to the SPL query?

BUILD *(2 Hours)*

BUILD BRIEF

Create a Detection Rule Library Document — your team's reference for all SIEM detection rules.

Required format for each rule:

- Rule Name, MITRE ATT&CK; Technique ID, Attack Description, SPL Query (exact, tested), True Positive Example (what it detects), False Positive Risk, Tuning Recommendations, Severity (Critical/High/Medium/Low), Escalation Action

Minimum 8 rules covering:

- SQL injection in URL parameters, XSS in URL parameters, SSH brute force (>10 failures), Successful login after brute force, Directory traversal (../../), Web scanner detection (Nikto/Gobuster signatures), High 4xx error rate from single IP, Privilege escalation via sudo

Format:

- Professional reference document. Each rule on its own page for operational use.

DOCUMENT

Writing Prompt:

Write a Detection Engineering Best Practices Guide for your SOC.

- Required: What makes a good detection rule (specific criteria), the detection engineering lifecycle (build → test → tune → deploy → monitor), how to measure rule quality (true positive rate, false positive rate), how to prioritise which attacks to detect first (use MITRE ATT&CK; top techniques)
- Include: a specific example of a bad rule, why it's bad, and how you would improve it

DAY 4

Wazuh HIDS/XDR and AI-Powered Anomaly Detection

LEARN *(1 Hour)*

LEARNING OBJECTIVES

1. Explain how Wazuh HIDS works and how it differs from Splunk

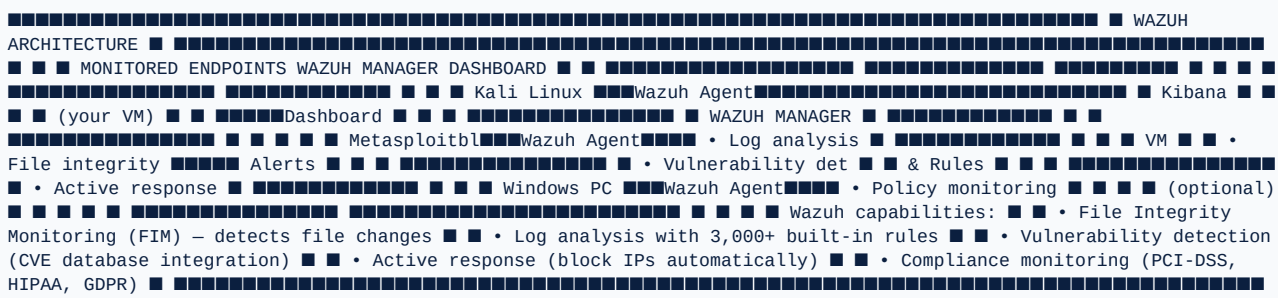
2. Install Wazuh using Docker and connect a monitored endpoint

3. Understand how AI/ML powers modern anomaly detection in SIEM

HIDS (HOST-BASED INTRUSION DETECTION SYSTEM)

monitors a single host or endpoint for signs of intrusion by analysing system calls, file changes, log entries, and network connections originating from that host — as opposed to network-based IDS which monitors traffic between hosts.

Wazuh Architecture



Wazuh vs Splunk — When to Use Each

Use Case	Splunk	Wazuh
Learning SPL for SOC jobs	✓ Primary choice	✗ Different query language
Production SIEM on tight budget	✗ Limited free version	✓ Fully free, unlimited
File integrity monitoring	✗ Requires additional config	✓ Built-in, easy to enable
Vulnerability detection	✗ Not built-in	✓ CVE database integration
Active response (auto-block)	✗ Not built-in	✓ Built-in active response
Compliance reporting	✗ Requires paid add-ons	✓ PCI-DSS, HIPAA, GDPR built-in
Nigerian SME / startup SOC	✓ Good for learning	✓ Better for production

AI and ML in SIEM — How Anomaly Detection Works

Traditional SIEM: rule-based — you define what looks suspicious. AI-powered SIEM: learns what 'normal' looks like and flags deviations.

- Baseline learning: AI models normal behaviour (login times, data volumes, commands run) over 2–4 weeks
- Anomaly detection: flags deviations from baseline (login at 3am, 10x normal data transfer, new command never seen before)
- User and Entity Behaviour Analytics (UEBA): AI tracks user behaviour patterns — detects compromised accounts even without known IoCs
- ML-powered threat hunting: AI surfaces suspicious patterns human analysts might miss across millions of events

AI/ML Technique	What It Detects	Example
Statistical anomaly	Events outside normal distribution	Login at 3am (user always logs in 8am-6pm)
Clustering	Grouping similar events — outliers stand out	Single user accessing 100 files/hour when avg is 5
Time series analysis	Unusual patterns in time-sequenced data	Spike in outbound traffic at midnight
NLP on log text	Malicious commands in free-text fields	PowerShell -enc (base64 encoded command)
Graph analysis	Unusual relationships between entities	User accessing systems never accessed before

AI-Powered SIEM Tools

- Microsoft Sentinel: cloud-native SIEM with built-in ML — uses Azure AI for UEBA and threat intelligence
- Splunk MLTK (Machine Learning Toolkit): add-on for Splunk that enables ML models within SPL
- Darktrace: AI-native network detection and response — learns 'digital DNA' of the organisation
- IBM QRadar: enterprise SIEM with Watson AI for threat investigation assistance
- Using Claude/ChatGPT for SIEM: paste log excerpts, ask for anomaly identification, generate detection rule suggestions

WHY THIS MATTERS

The 2020 SolarWinds breach went undetected for 9 months across 18,000 organisations including US government agencies. The attackers specifically designed their behaviour to look 'normal' — mimicking legitimate SolarWinds update traffic. AI-powered UEBA would have flagged the subtle pattern deviations that rule-based SIEM missed. This breach is the primary reason enterprises are investing heavily in ML-powered SIEM.

PRACTICE *(2 Hours)*

LAB 6.4.1 — Deploy Wazuh using Docker

Objective: Install Wazuh Manager and connect your Kali Linux VM as a monitored endpoint.

Prerequisites: Docker installed on Kali; internet connection

Step 1 — Install Docker (if not already installed)

BASH

```
sudo apt update && sudo apt install docker.io docker-compose -y sudo systemctl start docker && sudo systemctl enable docker sudo usermod -aG docker $USER # Then logout/login
```

Step 2 — Deploy Wazuh with Docker Compose

BASH

```
# Clone Wazuh Docker repository git clone https://github.com/wazuh/wazuh-docker.git -b v4.7.3 cd wazuh-docker/single-node # Start Wazuh (takes 3-5 minutes first time) docker-compose up -d # Check status docker-compose ps
```

Step 3 — Access Wazuh Dashboard

Navigate to: https://localhost (accept the self-signed certificate). Default credentials: admin / SecretPassword (check docker-compose.yml for current default). You should see the Wazuh dashboard with 0 agents connected.

Step 4 — Install Wazuh Agent on Kali

BASH

```
# Download and install Wazuh agent curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh.gpg echo 'deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main' | sudo tee /etc/apt/sources.list.d/wazuh.list sudo apt update && sudo apt install wazuh-agent # Configure agent to connect to manager sudo sed -i 's/MANAGER_IP/localhost/g' /var/ossec/etc/ossec.conf # Start agent sudo systemctl start wazuh-agent
```

Step 5 — Verify Agent Connected

In Wazuh Dashboard → Agents — you should see your Kali VM as an active agent. Click on it to see: security events, vulnerability scan results, file integrity monitoring status.

LAB 6.4.2 — AI-Assisted Log Analysis

Objective: Use Claude to analyse log patterns and generate detection rules.

Prerequisites: Splunk running with dvwa index; browser access to AI assistant

Step 1 — Export Suspicious Events from Splunk

Run this SPL query and export results as CSV:

SPL

```
index=dvwa | head 50 | table _time, clientip, method, uri_path, status, bytes
```

Step 2 — AI Analysis

PROMPT

```
I am a security analyst. Below are 50 Apache web server log events from my lab. Analyse them and tell me: 1. Which events look suspicious and why 2. What attack patterns you can identify 3. Suggest 3 Splunk SPL detection rules for the attack patterns you found 4. Flag any false positives in your suggested rules [paste your log data here — use only your own lab data, never real client logs]
```

Compare the AI's detection rule suggestions to the ones you wrote in Day 3. What did the AI find that you missed? What did you find that the AI missed?

■ ANALYSE THIS

The AI suggested a rule to detect 'unusual HTTP methods' — specifically DELETE and PUT on your DVWA. Your actual DVWA only uses GET and POST. Is this a good detection rule for your environment? How does 'baselining normal behaviour' improve detection rule quality compared to generic rules?

BUILD *(2 Hours)*

BUILD BRIEF

Create a SIEM Tool Comparison Report for your security team.

Scenario:

- Your team needs to choose a SIEM for a 50-person Nigerian fintech startup. Budget: minimal. Compliance requirement: NDPA 2023 and CBN guidelines.

Compare Splunk Free vs Wazuh vs Microsoft Sentinel (free tier) across:

- Cost (initial + ongoing), Data ingestion limits, Ease of deployment, Built-in detection rules, Compliance support (NDPA/CBN alignment), Skill requirements for team, Scalability

Required output:

- Comparison table with scores (1-5) for each criterion and justification
- Executive recommendation: which tool and why
- Implementation roadmap: 30-day setup plan for the chosen tool

Format:

- Professional consulting report format

DOCUMENT

Writing Prompt:

Write an AI in SOC: Opportunities and Risks report for your organisation's security leadership.

- Required: 3 specific opportunities AI brings to SOC operations (with examples), 3 specific risks of AI in security monitoring (hallucinations, bias, adversarial attacks on ML models), your recommendation: which AI-powered SIEM features to adopt first and why
- Tone: Balanced and evidence-based. Every claim supported by a real example or data point

DAY
5

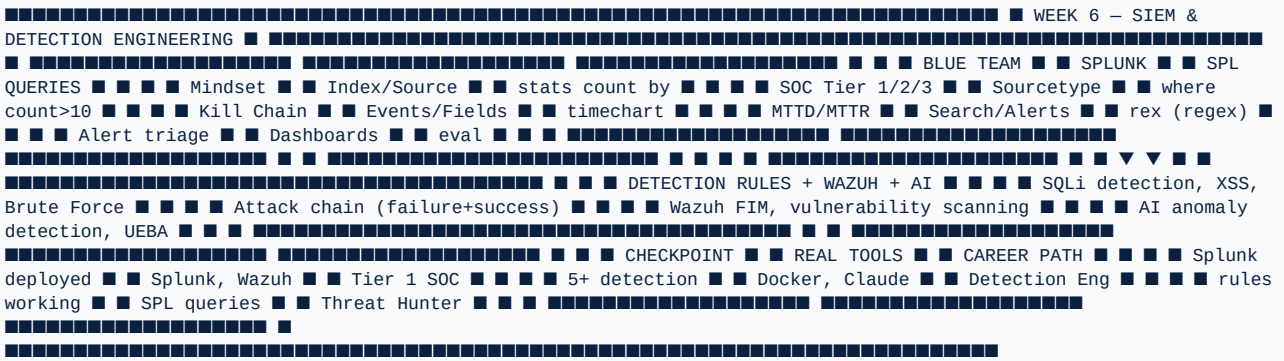
Review, Assessment & Portfolio (Checkpoint Project)

LEARN *(1 Hour) — Week Review*

LEARNING OBJECTIVES

1. Demonstrate all Week 6 skills without referencing notes
2. Complete the Checkpoint Project — your most significant portfolio piece so far
3. Plan the second half of the course

Week 6 Mind Map — SIEM & Detection Engineering



PRACTICE *(2 Hours) — Skills Assessment Labs*

LAB 6.5.1 — SPL Rapid-Fire Challenge

Objective: Write SPL queries from memory in 30 minutes without notes.

Prerequisites: Splunk running with dvwa index; no notes

Write SPL queries for each requirement. Every query must run and return results.

1. Show the top 10 IP addresses by number of requests in the last 24 hours
2. Find all HTTP 500 errors in the last hour
3. Count events per hour for today (timechart)
4. Find all events containing the string 'union' in the URL (case insensitive)
5. Show all failed SSH logins, count by source IP, only show IPs with more than 5 failures
6. Correlate: find IPs that had both failed logins AND SQL injection attempts

LAB 6.5.2 — Detection Rule Accuracy Test

Objective: Evaluate the quality of your detection rules by testing against known traffic.

Prerequisites: Splunk with dvwa index; your detection rule library from Day 3

For each detection rule you built, run it against your log data and calculate:

- True Positives: known attack events correctly identified
- False Positives: legitimate events incorrectly flagged
- False Negatives: known attacks that the rule missed

Based on your analysis: which rule needs tuning and how would you tune it?

BUILD *(2 Hours) — Checkpoint Project Submission*

This is your Checkpoint Project — the most significant deliverable of Phase 2. You must demonstrate a working SIEM with real detection capabilities. This proves you can operate as a Tier 1/2 SOC analyst.

Deliverables:

- Splunk Deployment Evidence: screenshot of Splunk running, data sources configured, event count by sourcetype
- Data Ingestion Report: sources ingested (Apache, auth.log), volume (events per day), time range covered
- Detection Rule 1 — SQL Injection: SPL query, test results (true positive confirmed), screenshot of detection firing, alert configuration
- Detection Rule 2 — Brute Force SSH: SPL query, test results, alert configuration, false positive analysis
- Detection Rule 3 — Attack Chain (Brute Force → Success): SPL query, test results, severity justification
- Detection Rule 4 — XSS: SPL query, test results, false positive tuning notes
- Detection Rule 5 — Your Choice: any additional attack pattern you detected, with full documentation
- Security Dashboard: screenshot of your Splunk dashboard with at least 4 panels running in real-time
- Wazuh Integration (Bonus): screenshot of Wazuh agent connected, FIM alert generated, vulnerability scan results
- AI-Assisted Analysis (Bonus): screenshot of Claude/AI analysis of your logs, comparison of AI suggestions vs your manual rules
- Lessons Learned: what you would do differently in a production SOC vs a lab environment

Tools: Splunk Free · Wazuh · Docker · SPL · Claude AI (documented) · All Week 3–6 skills

Checkpoint Project — Submission Details

- Due: End of Week 6 (today)
- Format: PDF report + Splunk dashboard screenshot + SPL query document

- Evaluation: Splunk deployment (20%), Detection rule quality (40%), Dashboard (15%), Documentation (15%), Bonus items (10%)
- This is 15% of your total course grade — treat it as seriously as a real client deliverable

DOCUMENT *(1 Hour) — Week Summary*

DOCUMENT

Writing Prompt:

Write your Week 6 learning reflection. Length: 300–400 words.

- The most powerful SPL query you wrote and what it detects
- How does writing detection rules change how you think about attacking systems?
- What surprised you most about SIEM implementation — what was harder or easier than expected?
- What would your Splunk dashboard look like if you were monitoring a Nigerian bank's 10,000 employee network?
- Prediction: which Week 5 attack (SQLi, XSS, or brute force) is hardest to detect and why?

Weekly Checklist (Student-Facing)

- ■ I can explain the difference between a Tier 1, 2, and 3 SOC analyst without notes
- ■ I can map all 7 Kill Chain phases to a defensive control and detection opportunity
- ■ Splunk is installed and ingesting DVWA Apache logs and auth.log
- ■ I have written and tested a SQL injection detection rule in SPL
- ■ I have written and tested a brute force detection rule in SPL
- ■ I have built a Splunk dashboard with at least 4 security monitoring panels
- ■ I have created at least one Splunk alert that fires when an attack is detected
- ■ I have deployed Wazuh using Docker and connected at least one agent
- ■ I have used AI to analyse log data and compared the output to my manual analysis
- ■ My Checkpoint Project is complete and submitted

Skills Verification

Skill	How to Demonstrate It	Pass Standard
SOC knowledge	Explain Tier 1/2/3 roles and MTTD/MTTR without notes	Correct definitions + realistic metric examples
Splunk deployment	Show Splunk running with events from dvwa index	At least 1,000 events indexed from two sources
SPL basics	Write 3 SPL queries from memory: count by IP, filter by status, timechart	All 3 queries run and return results
SQLi detection	Show SPL rule that detects SQLi payloads in URL	Rule fires on your Week 5 SQLi attacks
Brute force detection	Show SPL rule detecting >10 SSH failures from same IP	Rule fires on your generated attack traffic
Dashboard	Show Splunk dashboard with 4 real-time panels	All panels displaying live data
Wazuh	Show Wazuh dashboard with at least 1 active agent	Agent connected; FIM and log analysis active

Self-Assessment Quiz

Instructions

- No notes — answer from memory.
- Write on separate sheet.
- Key in Instructor Notes.

Conceptual Questions

1. A SOC has MTTD of 197 days and MTTR of 69 days. What does this mean in plain English? What is the total exposure window? How would you prioritise reducing which metric first and why?
2. Explain the difference between a SIEM and an IDS. Which would detect a SQL injection in a web server log? Which would detect a port scan in network traffic? Which would detect both?
3. What is the difference between a true positive, false positive, false negative, and true negative in SIEM alerting? For a brute force detection rule, give a specific example of each.

SPL Questions

4. Write a SPL query from memory to: find all events in the dvwa index with HTTP status 200, where the URI contains the word 'admin', grouped by clientip with a count, sorted descending.
5. Your SQL injection detection rule is generating 200 false positives per day because legitimate users search for products with SQL keywords in the name (e.g., 'SELECT Coffee'). Write an improved SPL query that reduces false positives while maintaining detection of actual SQL injection.

Scenario Questions

6. At 2:47 AM, your Splunk alert fires: '47 failed SSH logins from IP 185.220.101.5 in the last 10 minutes, followed by 1 successful login.' You are the Tier 1 analyst. Describe exactly what you do in the next 15 minutes (step-by-step actions, what you check, what you document, when you escalate).
7. Your Wazuh dashboard shows a File Integrity Monitoring (FIM) alert: /etc/passwd was modified at 3:15 AM by user 'www-data'. Why is this suspicious? What does it likely indicate? What is your immediate response?

Nigerian Context

8. A Nigerian bank's SIEM fires an alert: a user in the Lagos branch accessed 500 customer account records in 10 minutes at 11 PM. Under the NDPA 2023: is this a reportable incident? What must the bank do in the next 72 hours if personal data was accessed by an unauthorised person? What SIEM rule would have prevented this from going undetected?

Additional Resources

Splunk Resources

- Splunk Free Download: https://www.splunk.com/en_us/download.html
- Splunk Fundamentals 1 (Free Training): https://www.splunk.com/en_us/training/free-courses.html
- Splunk Documentation — SPL Reference: https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/
- Splunk Security Essentials App (Free): https://splunkbase.splunk.com/app/3435

Wazuh Resources

- Wazuh Documentation: https://documentation.wazuh.com
- Wazuh Docker Quick Start: https://documentation.wazuh.com/current/deployment-options/docker/wazuh-container.html
- Wazuh SIEM Rules Reference: https://documentation.wazuh.com/current/user-manual/ruleset/

SOC and Detection Engineering

- MITRE ATT&CK; Navigator (map your detection coverage): https://mitre-attack.github.io/attack-navigator/
- Sigma Rules (vendor-agnostic detection rules): https://github.com/SigmaHQ/sigma
- TryHackMe — Splunk 101 Room (Free): https://tryhackme.com/room/splunk101
- TryHackMe — Splunk 2 Room: https://tryhackme.com/room/splunk2gcd5
- Blue Team Labs Online (Free labs): https://blueteamlabs.online

AI in Security

- Microsoft Sentinel (cloud SIEM with AI): https://azure.microsoft.com/en-us/products/microsoft-sentinel
- Darktrace (AI NDR): https://darktrace.com
- IBM QRadar SIEM: https://www.ibm.com/products/qradar-siem

Looking Ahead — Week 7

Preview — What's Coming in Week 7

- Intrusion Detection Systems (IDS) — Snort configuration and rule writing
- Network traffic analysis — capturing and analysing attacks at the packet level
- IPS (Intrusion Prevention System) — automatically blocking detected attacks
- Integrating IDS alerts with your Splunk SIEM for unified detection
- Your Week 4 Nmap scan and Week 5 web attacks will generate IDS alerts — you will detect them

What to Review Before Week 7

- Your Week 2 OSI model and TCP/IP knowledge — IDS operates at network layers
- Your Week 3 Wireshark skills — IDS rule writing requires understanding packet structure
- Your SPL skills from Week 6 — you will correlate IDS alerts with SIEM events

What to Set Up Before Week 7

- Install Snort on Kali: sudo apt install snort
- Verify Snort version: snort --version
- Download community Snort rules: https://www.snort.org/downloads#rules
- Keep DVWA running — Week 7 IDS rules will detect Week 5 attack traffic

Checkpoint Project Reminder

- Your Checkpoint Project is due today — submit your PDF report before end of session
- If you have not deployed Wazuh yet, focus on the Splunk deliverables first — they carry more marks
- Keep your Splunk dashboard running — you will build on it in Week 7